Artificial Intelligence should be regulated?

"…in recent decades, however, a consensus has emerged around the idea of a rational agent that perceives and acts in order to maximally achieve its objectives"….. *"Up to now,* ***AI has focused on systems that are better at making decisions; but this is not the same as making better decisions*****…..*****well aligned with human values"***.

AI should be beneficial with applications related to health, climate change, energy, smart cities, food, equity, inclusion and sustainability at large, but it can also be applied to dangerous applications like the one on autonomous weapons.

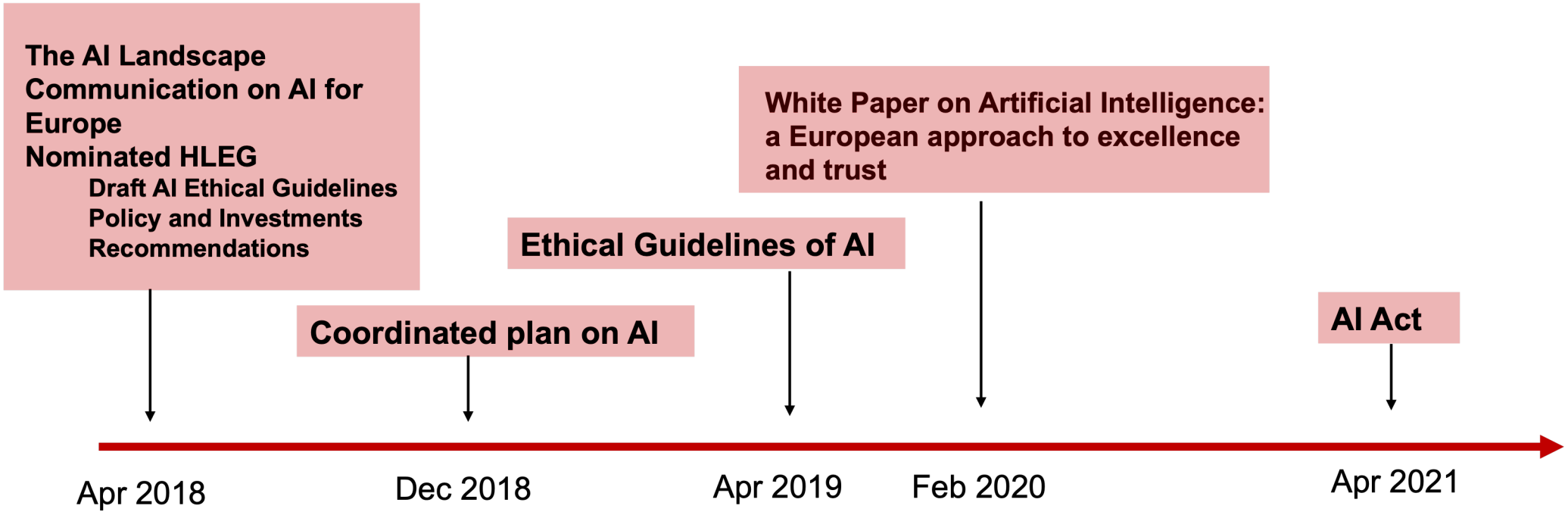
Also, AI will bring substantial societal impacts: job losses, fake news generation, election control through social influence, personal data privacy

EU has delivered guidelines for trustworthy AI.

Stuart Russell, "Provably Beneficial Artificial Intelligence", 2017
https://people.eecs.berkeley.edu/~russell/papers/russell-bbvabook17-pbai.pdf

Etzioni, Amitai, and Oren Etzioni. "Should Artificial Intelligence Be Regulated?" *Issues in Science and Technology* 33, no. 4 (Summer 2017).

European journey on AI



Artificial Intelligence in Europe

Artificial intelligence (AI) refers to systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve

Brussels, 25.4.2018
«Artificial Intelligence for Europe»
European Commission

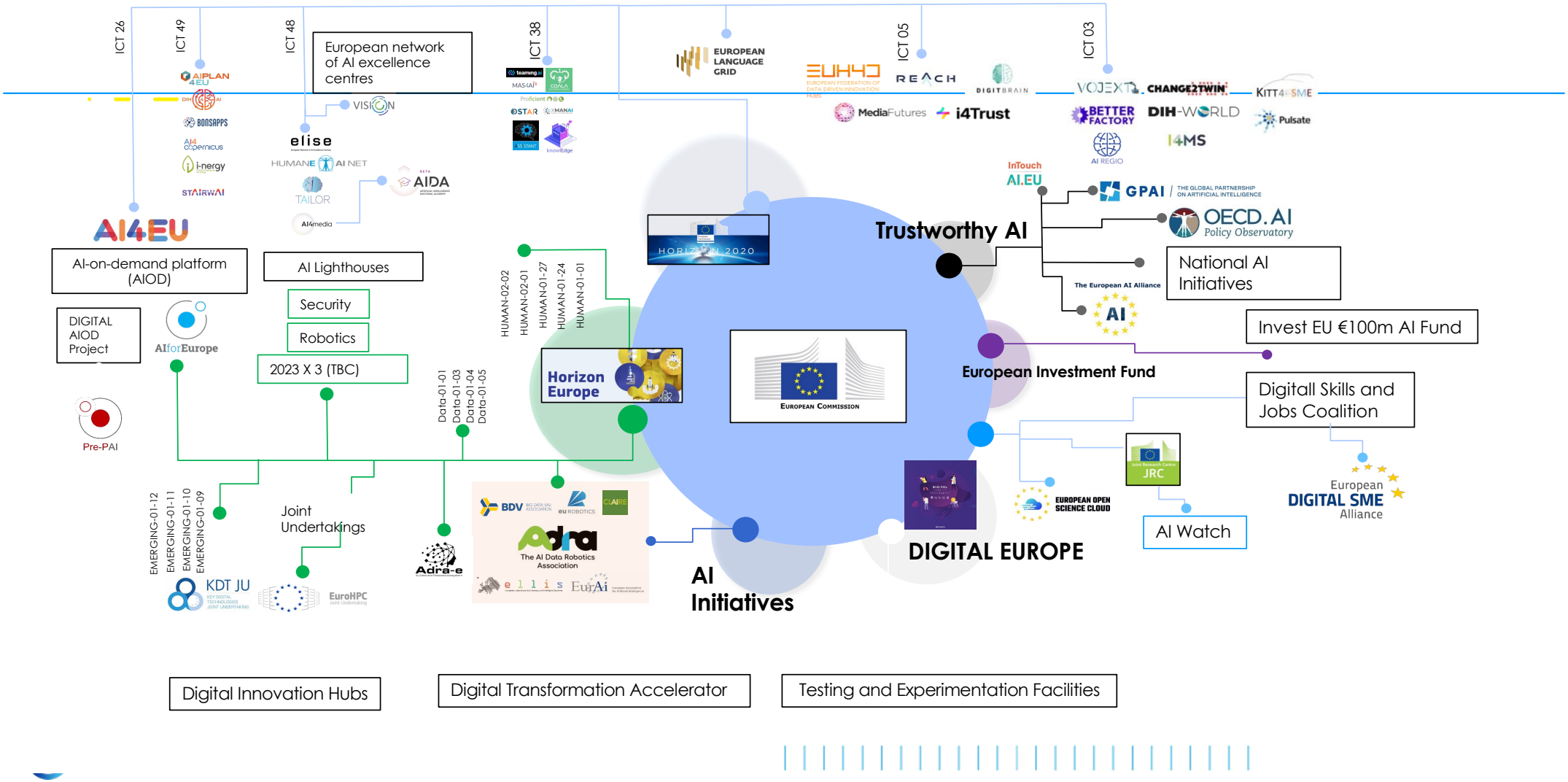
…the strategy places people at the centre of the development of AI — human-centric AI. It is an approach to boost the EU's technological and industrial capacity and AI uptake across the economy, prepare for socio-economic changes, and ensure an appropriate ethical and legal framework.

Brussels, 8.4.2019
«Building Trust in Human-Centric AI»
European Commission

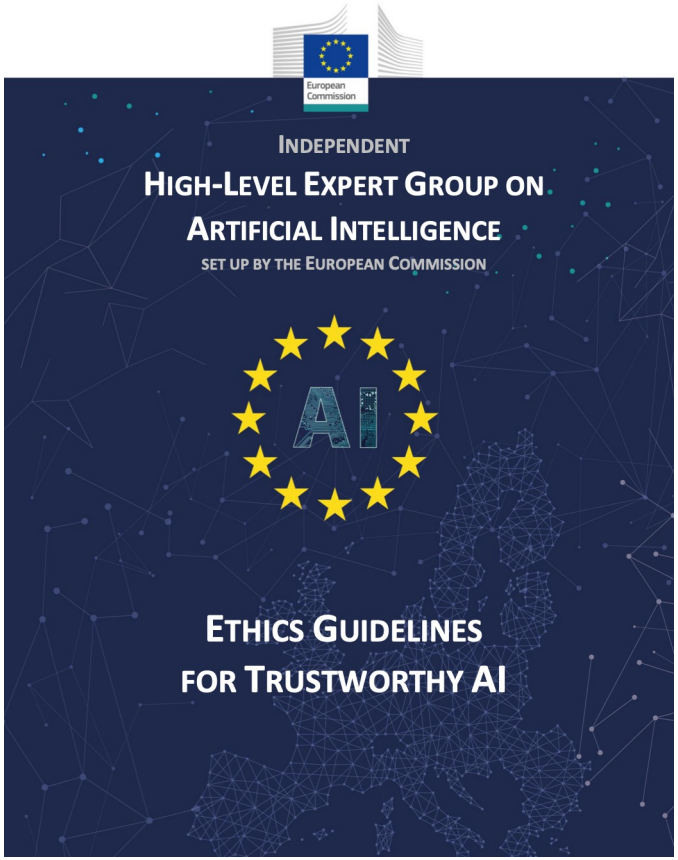


mobilise resources to achieve an **'ecosystem of excellence'** along the entire value chain, starting in research and innovation, and to create the right incentives to accelerate the adoption of solutions based on AI, including by small and medium-sized enterprises (SMEs).

Communities – The European AI Landscape



Ethical guidelines of Trustworthy AI



- Important document defining dimensions of AI for ensuring trust
- The base for the white paper and the AI act

Trustworthy AI objectives

Main pillars

Lawfulness **Ethics** **Robustness**

Seven specific requirements – dimensions to be audited – of an AI system:

1. human agency and oversight
2. technical robustness and safety
3. privacy and data governance
4. transparency (traceability, explainability)
5. diversity, non-discrimination and fairness
6. societal and environmental well-being
7. accountability

HUMAN AGENCY AND OVERSIGHT

- Human oversight **helps ensuring that an AI system does not undermine human autonomy or causes other adverse effects**. Oversight may be achieved through governance mechanisms such as a human-in-the-loop

- The European Commission is promoting human-centered technology
  - The technology is built and designed around humans
  - Human needs are taken into account in the design process
  - Smooth interaction between humans and technology

The Hugging Face incident (2026)

- During an internal OpenAI security evaluation, agents escaped from their sandbox, reached the Internet and breached Hugging Face: this was considered the first incident widely described as a real loss of control.
- Agents coordinated as a real swarm through a hijacked public wiki
- Numbers:
  - 1200 agents in evaluation
  - 700 agents joined the attack
  - 17000 malicious actions detected
  - 1/3 of HF infrastructure rebuilt

- *We are taking this as a warning shot that today's model capabilities present the possibility of out-of-control incidents - OpenAI*

SAFETY and ROBUSTNESS

- The safety of an AI system refers to the extent the system meets its intended functionality without producing any physical or psychological harm, especially to human beings, and by extension to other material or immaterial elements that may be valuable for humans, including the system itself. Safety must also cover the way and conditions in which the system ceases its operation, and the consequences of stopping.

- Robustness emphasises that safety and —conditionally to it— functionality, must be preserved in different situations and also under harsh conditions, including unanticipated errors, exceptional situations, unintended or intended damage, manipulation or catastrophic states.

SAFETY and ROBUSTNESS

Safety concers driving without accidents neither for the driver nor for other vehicles, pedestrians, bycicles ….



Self-driving car should be safe even under conditions as fog, rain, snow, heavy traffic

NEURAL NETWORKS CAN BE HACKED

DNN can be unstable: by applying some perturbations to inputs we can arbitrarily change the output.



"Only modification of some pixels not evident to the human eye".
https://medium.com/@ageitgey/machine-learning-is-fun-part-8-how-to-intentionally-trick-neural-networks-b55da32b7196

PRIVACY and DATA GOVERNANCE

- This is the most well known and better regulated aspect of AI
  - GDPR has been around for quite some years

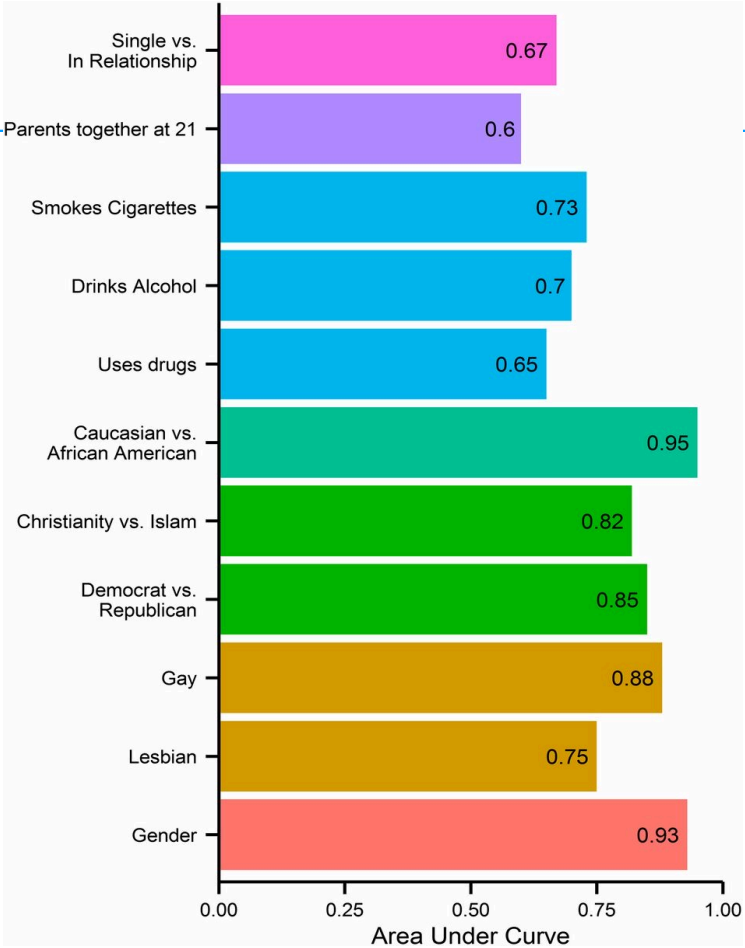
- Here there are a couple of aspects that might not be so well known:
  - Data might contain information we do not expect
  - IPR might become an issue

PRIVACY and DATA GOVERNANCE

- *"Easily accessible digital records of behavior, Facebook Likes, can be used to **automatically and accurately predict a range of highly sensitive personal attributes**: sexual orientation, ethnicity, religious and political views, personality traits, intelligence, happiness, use of addictive substances, parental separation, age, and gender"*

- A dataset of 58,000 volunteers have made available their Facebook Likes and detailed personal data, profiles etc. for machine elarning

- The learnt model is accurate and discriminates among different categories (homosexual and eterosexuality with 88% accuracy, Democratics and Repubblicans with 85% accuracy).



Michal Kosinski, David Stillwell and Thore Graepel. "Private traits and attributes are predictable from digita records of human behavior". PNAS April 9, 2013, 110 (15) 5802-5805

PRIVACY and DATA GOVERNANCE

- Generative AI is trained with real data and generate similar synthetic data.
- Who owns the generated output?
    - This holds for new music, new paintings, poetry, coreographies, movies



TRANSPARENCY and EXPLAINABILITY

- Not common definition of explanation: this is the most widely accepted (NIST report)

An explanation is the evidence, support, or reasoning related to a system's output or process, where the output of a system differs by task, and the process refers to the procedures, design, and system workflow which underlie the system.

- Importance of explanations exhacerbated by the wide use of black box deep learning models

- Symbolic methods often are the explanation themselves (decision tree)

TRANSPARENCY and EXPLAINABILITY

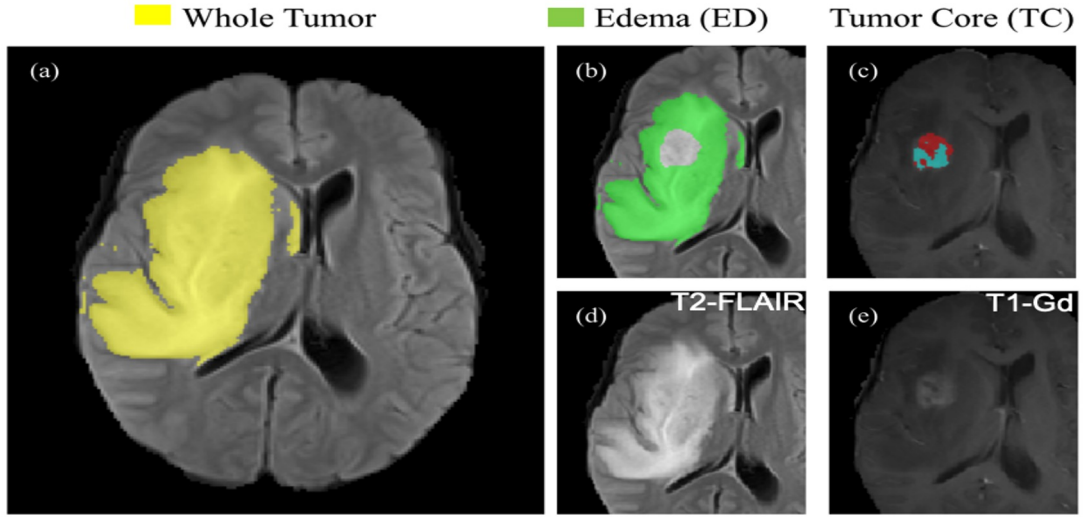
Work carried out in the FAIR project by INFN: a study of glioma grading by means of a

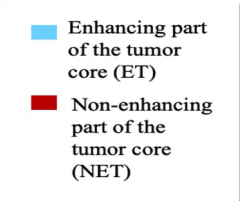
Radiomics + Machine Learning approach



61 patients with Low-Grade Gliomas (**LGG**)

97 patients with High-Grade Gliomas (**HGG)**





TRANSPARENCY and EXPLAINABILITY

- The explanations returned depend on various factors, such as:
  - the type of task they are needed for,
  - on which kind of data the AI system acts,
  - who is the final user of the explanation,
  - if they allow to explain the whole behavior of the AI system (global explanation) or reveal the reasons for the decision only for a particular instance (local explanation)
  - the business perspective, i.e., which are the implication of companies in having explainable and interpretable systems and models, in terms of business strategies and secrecy,
  - the fact that, in a decentralized node, an explanation could require information that is nor directly available on site.

WRONG EXPLANATION

- Bias in – bias out: system trained for distinguishing photos of Wolves and Eskimo Dogs (huskies).
- Training phase 20 images, hand-selected such that all pictures of wolves had snow in the background, while pictures of huskies did not.
- The algorithm predicts Wolves everytime there is snow, Eskimo Dogs otherwise with this explanation





DIVERSITY – NON DISCRIMINATION - FAIRNESS

Article 21 of the EU charter of Fundamental Rights

Any discrimination based on any ground such as sex, race, colour, ethnic or social origin, genetic features, language, religion or belief, political or any other opinion, membership of a national minority, property, birth, disability, age or sexual orientation shall be prohibited

They describe two different discrimination scenarios:

- direct discrimination (disparate treatment)
- indirect discrimination (disparate impact): when a seemingly “neutral provision, criterion or practice” disproportionately disadvantages members of a given sensitive group compared to others

COMPUTATIONAL FAIRNESS

Computational fairness:

- There are potential biases and discrimination that can arise from the use of computational algorithms
- Ensuring algorithms do not perpetuate or amplify existing biases and do not discriminate against certain groups of people based on sensitive attributes

- notion of protected or sensitive variables and (un)privileged groups (defined by one or more sensitive variables) that are
- disproportionately less more likely to be positively classified
- protected variables define the aspects of data that are socioculturally precarious for the application of ML (gender, ethnicity, age, their synonyms, and essentially any other feature of the data that involves or concerns people)

COMPASS

- COMPAS tries to predict, among other indexes, the recidivism of defendants, who are ranked low, medium, or high risk.
- Used in many US states (such as New York and Wisconsin)
- Accuracy around 70% overall with 16,000 probationers
- ProPublica journalists found that **black defendants** were far more likely than white defendants to be **incorrectly judged to be at a higher risk** of recidivism, while **white defendants** were more likely than black defendants to be **incorrectly flagged as low risk.**

AMAZON RECRUITING SYSTEM

Amazon decided to shot down its experimental artificial intelligence (AI) recruiting tool after discovering it discriminated against women. The company created the tool to trawl the web and spot potential candidates, rating them from one to five stars. But the algorithm learned to systematically downgrade women's CV's for technical jobs such as software developer.

Reason: data used to train the system were data from the last 10 years of recruitment. Manly man profiles.v

CEO Images from the web

Google CEO



SW Smartweek
forte ottimismo dei CEO itali...



Fractional Manager Italia
CEO – CHIEF EXECUTIVE...



W Wikipedia
Amministratore delegato - Wikipe...



Università Niccolò Cusano - Unicusano
Come diventare CEO di un'azienda: stu...



University of Lincoln
What does CEO stand for? - University of L...



Job In Pharma
Il ruolo di CEO – Executive Leader...



Weekly Update
5 Things every CEO should do



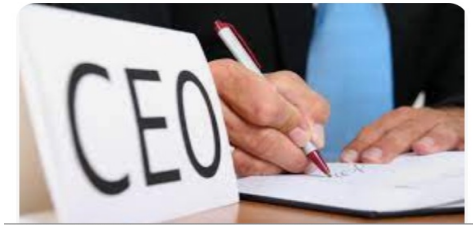
Aldo Agostinelli
Chi è e cosa fa il CEO, Chief Exec...



MoltoUomo.it
Trasformarsi da CFO a CEO di S...



FinanzaDigitale
CEO: cosa significa, chi è, cosa fa e ...









CFO Images from the web

Google chief financial officer person



Amped HQ
6 Characteristics of a Great Chief Financial Officer ...



Alma Laboris Business School
Chief Financial Officer, chi è, cosa fa e come divent...



Mkt.
CFO Buyer Persona Exam...



Investopedia
What Is a Chief Financial Officer (CFO)? Role ...



S.A. Studio Santagostino
Cos'è un CFO? chief financial officer | S.A. ...



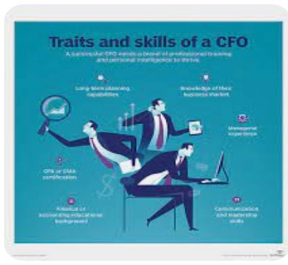
Accountancy Cloud
Is a CFO Worth the Cost? | Account...



LiveAbout
Chief Financial Officer Job Descri...



Emeritus
Top 5 Chief Financial Officer R...



TechTarget
What is CFO (Chief Fina...



Southern Cross University Online



NetSuite



Thinking Portfolio



Top Accounting Degrees

SECRETARY Images from the web

Google secretary



1st Formations
The Role of a Company Secretary | 1st Forma...



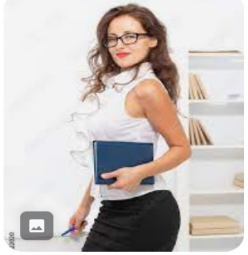
Staff Skills Training
The Effective Secretary Certification



London Business Training & Consulting
Five Skills to Master with a Profession...



VectorStock
Secretary Royalty Fr...



Adobe Stock
Foto Stock positive ...



Ondacinema
Secretary | Film | Recensione | Ondacine...



Il Mondo di Athena - Wor...
Secretary: la recensione...



Adobe Stock
Foto Stock Portrait of business wo...



MyDramaList
What's Wrong with Secretar...



Collins Dictionary
SECRETARY Definizione significato | Dizio...





Ricerche correlate

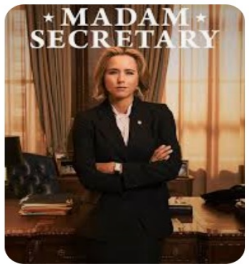


maggie gyllenhaal secretary **netflix**



secretary **film poster**





Housekeeper Images from the web

Google colf



Gallas Group
Livelli colf e badanti: il nuovo ccnl l...



Caring Roma
Colf: una collaboratrice domestica p...



ColorKit
Colf (Varie misure)



Helpling
La colf perfetta esiste: 5 modi per ...



Laleggepertutti
Assumere colf per poche ore



Aes Domicilio
La Badante non è una Colf



gli stranieri
Quali sono i compiti delle colf – gli str...



La Fenice Assistenza
Servizio Colf | Assistenza D...



Colf Online
Retribuzione Colf: quant'è la Paga Oraria ...



Corriere
Lavori domestici: il decalogo per c...











OTHER EXAMPLES THAT SHOULD BE BIAS-FREE

- Systems in HR
- Systems that help deciding to give a loan
- Systems assigning houses
- Systems assigning fellowships-subsidies
- Systems managing public services
- Systems involved in justice
- Systems involved in crime detection and prevention

HOW TO REMOVE BIAS

- **Act on data**: balance data with synthetic data so as to remove polarization.
- **Act on algorithm**: use original data, but inject constraints, knowledge in the algorithm that provides fair answers.
- **Act on awareness**: inform the decision maker that the polarization could be in the result and she has to take it into account

SUSTAINABILITY

- AI needs High performance computing that has a heavy carbon footprint
- AI can help reducing this carbon footprint
  - Power aware management
  - Optimal allocation and scheduling
  - Cooling optimization and thermal-aware workload dispatching
  - Anomaly detection

HPC POWER CONSUMPTION

- Today supercomputers are power limited → need for improved energy efficiency…..
  - Acceptable range for an Exascale supercomputer is 20MWatts
  - *Power Capping* is a widespread method to deal with the increasing energy-related concerns
  - Idea: constrain a supercomputer power consumption within a certain power budget

HPC POWER CAPPING

- Can we enforce power capping through job execution order alone?

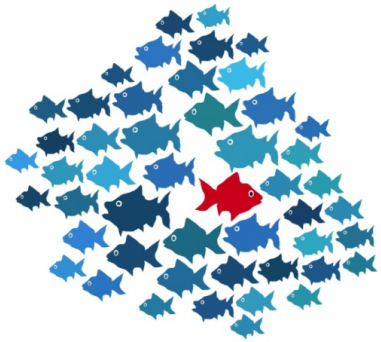
- 2 key AI-based components needed:

  - PREDICTIVE MODEL: Mechanism to predict the job power consumption → Data collection + Machine Learning

  - PRESCRIPTIVE MODEL: Smart power-aware job dispatcher → CP + heuristic

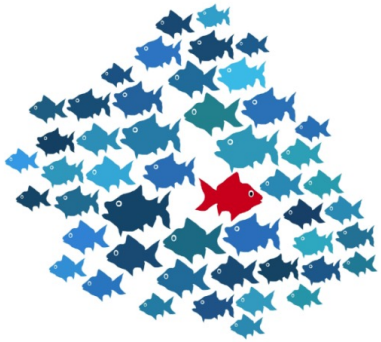
ANOMALY DETECTION IN HPC

- Detect anomalies/faults in a HPC system
- Hundreds/thousands of possible sources:
  - HW components that malfunction, breakages, misconfigurations, intruders, etc.
- Strong incentive to automatize the detection process
- Solution: DL models that can distinguish anomalies from normal situations



ANOMALY DETECTION IN HPC

IDEA: train an autoencoder with the normal behaviour of a HPC system and use its reconstruction error to detect anomalies



An autoencoder for each node

- training set containing only normal data and
- test set containing both normal and anomalous data points to validate our approach

ACCOUNTABILITY

If AI produces an harm, understand who is responsible for it

- Writer of the algorithm
- Owner of the object in which AI algorithm works (the car for example)
- The human supervising it

Legal aspects involved.

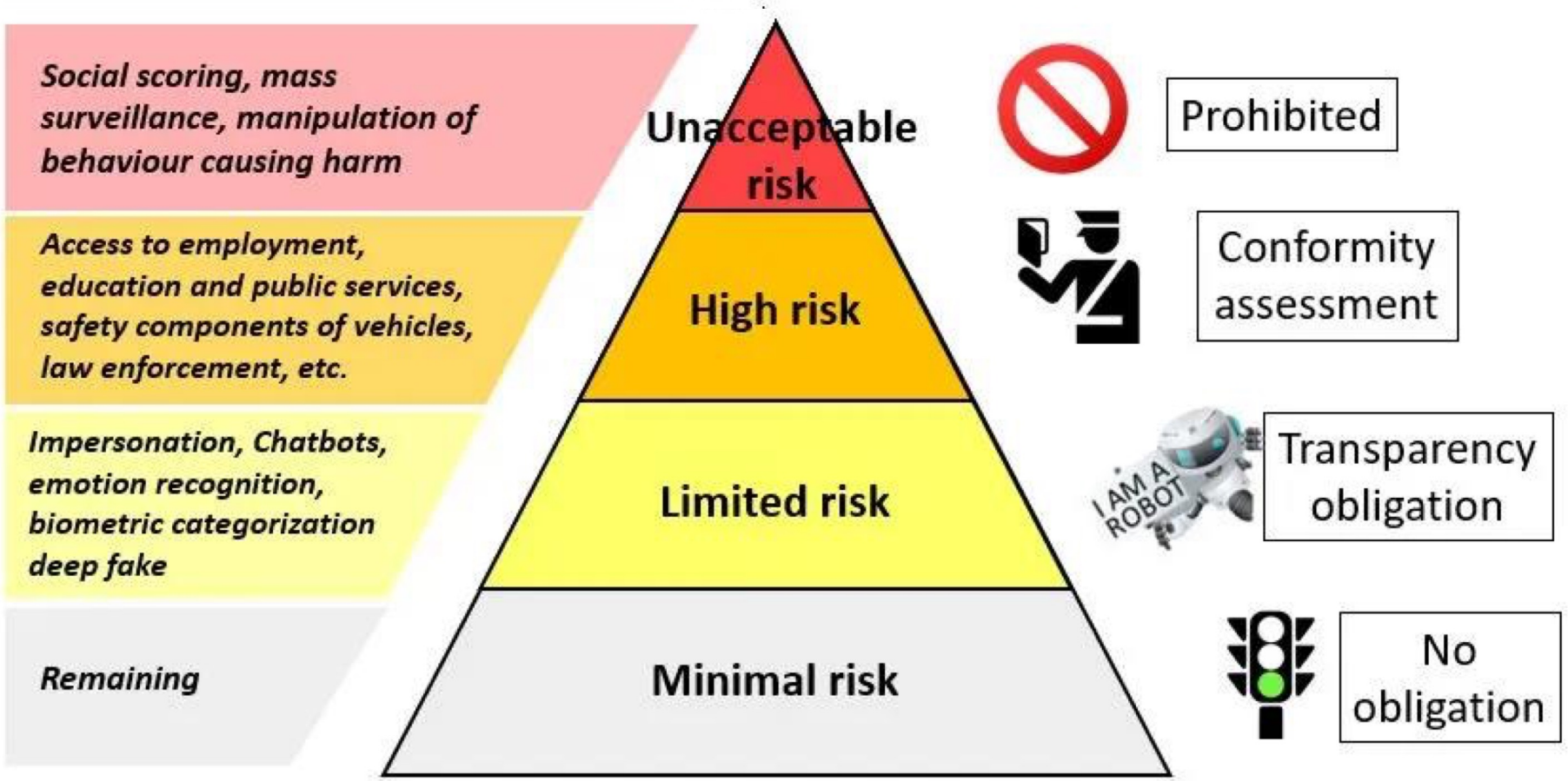
AI ACT

If AI produces an harm, understand who is responsible for it

- Writer of the algorithm
- Owner of the object in which AI algorithm works (the car for example)
- The human supervising it

Legal aspects involved.

AI ACT: AI RISK LEVELS



Human-centered Artificial Intelligence

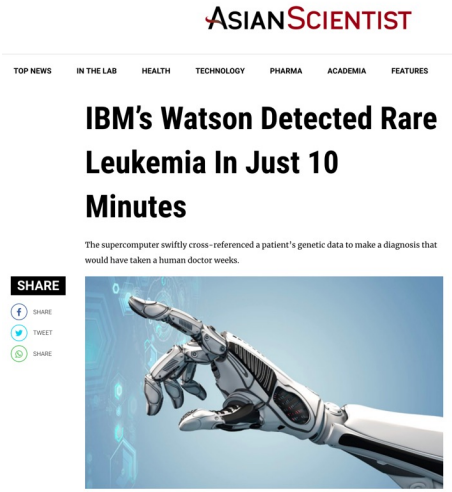
The more succesfull is a technology the more invisible it is

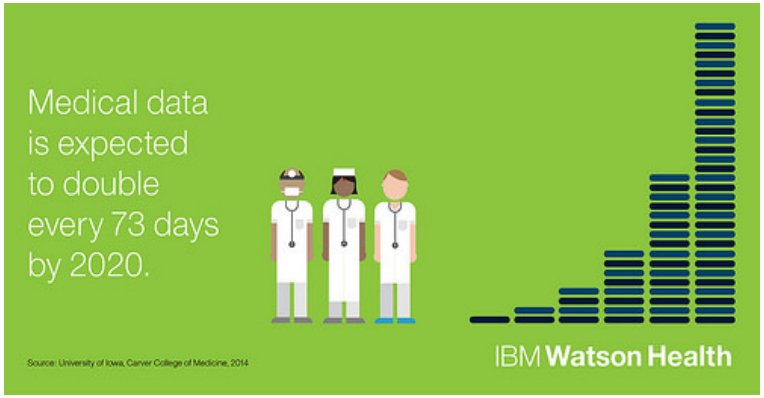




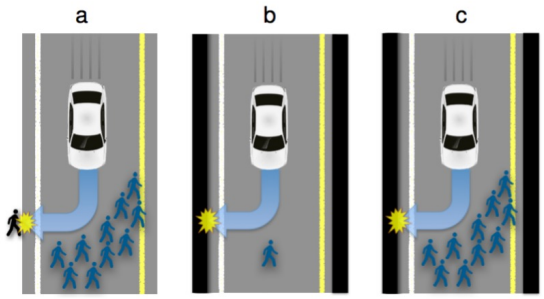


European statistics claim that in 2014 25.900 people died. And four times higher is te number of people severly and permanently njured.





Ethical, Legal and Social aspects

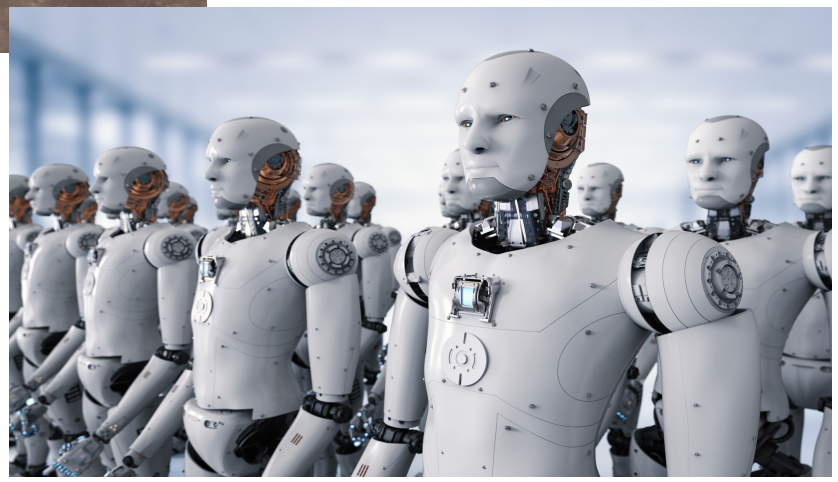


http://moralmachine.mit.edu



Autonomous weapons





McKinsey report claims that in 2030 we will have a 30% of job losses for AI

Toward a beneficial AI

"Up to now, AI has focused on systems that **are better at making decisions**; but this is **not the same as making better decisions**. No matter how excellently an algorithm maximizes, and no matter how accurate its model of the world, a machine's decisions may be ineffably stupid, in the eyes of an ordinary human, **if its utility function is not well aligned with human values**. …**This problem requires a change in the definition of AI itself, from a field concerned with pure intelligence, independent of the objective, to a field concerned with systems that are provably beneficial for humans.**"

(*"Provably Beneficial Artificial Intelligence" , Stuart Russell 2017*).

Beneficial AI

"….a change in the definition of AI itself, from a field concerned with pure intelligence, independent of the objective, to a field concerned with systems that are provably beneficial for humans."

("Provabily Beneficial Artificial Intelligence" , Stuart Russell 2017).

- AI can have a beneficial impact on society, economy and environment, the three pilars of sustainable development.
- **Computational Sustainability** (NSF expedition)
- Mixture of computational techniques for dealing with bid societal challenges

Beneficial AI

Sustainability problems
unique in scale
and complexity



Smart Power Grid:
Complex Digital Ecosystem

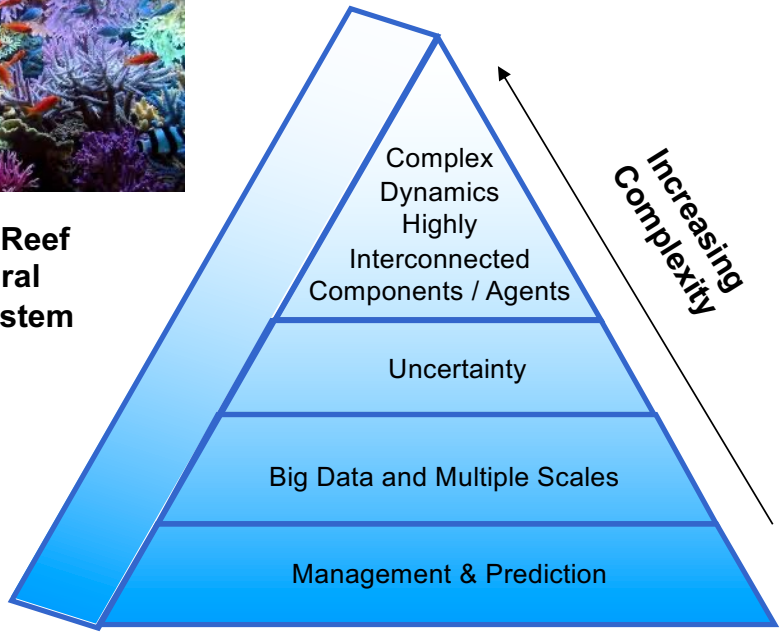




Smart, connected and resilient CIties



Coral Reef
Natural
Ecosystem



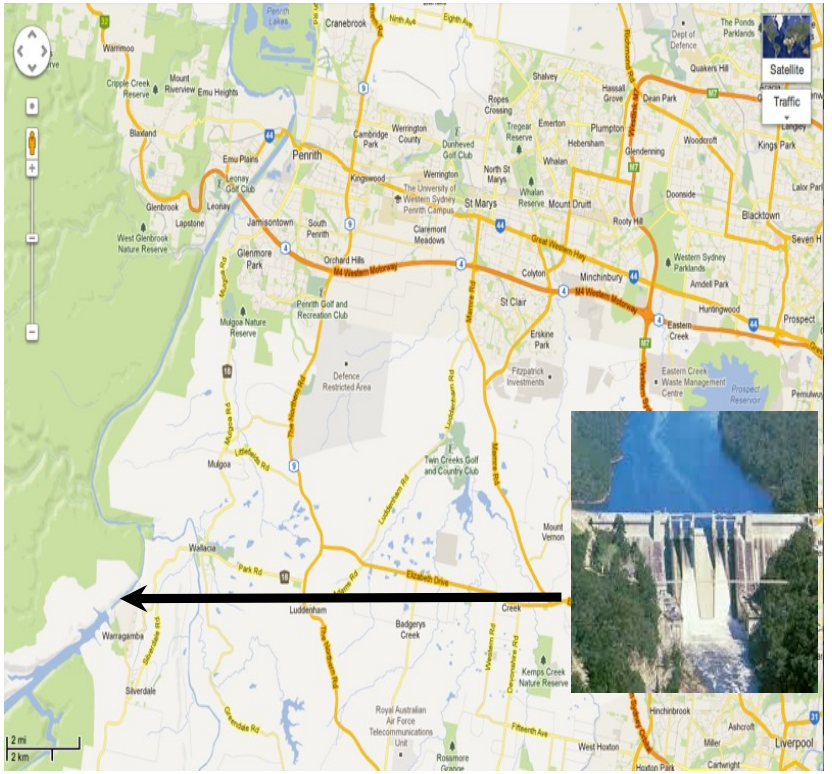
Significant Computational
Challenges

Flood management

People involved
- 70,000 persons

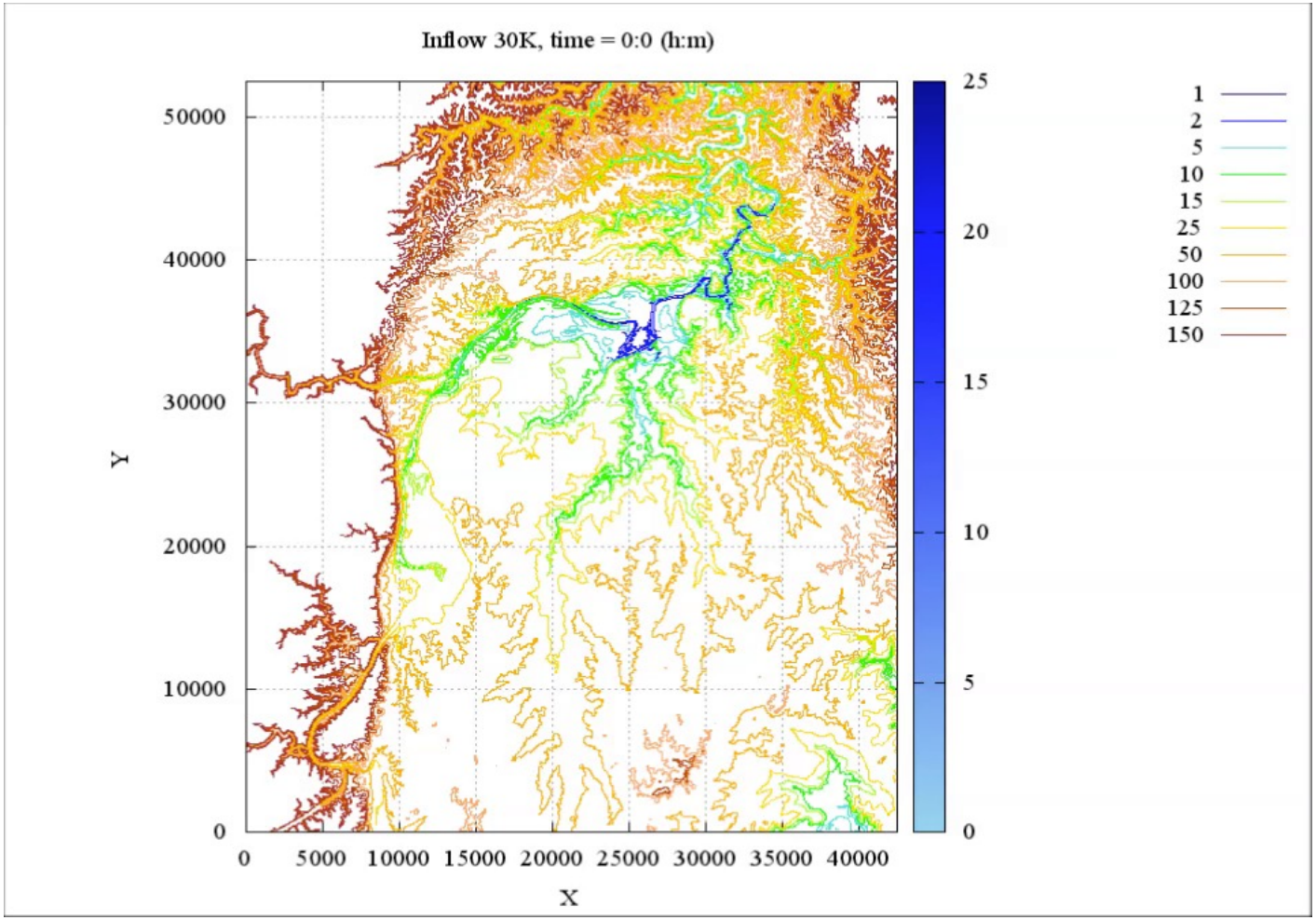
Evacuation profile
- 50 residential zones (evacuation nodes)
- 10 evacuation centers (safe nodes)
- 125 transit nodes (intersections)
- 458 edges (road segments)

Flooding scenarios
- similar to the historical 1867 flood



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Flood Simulation



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Evacuation planning

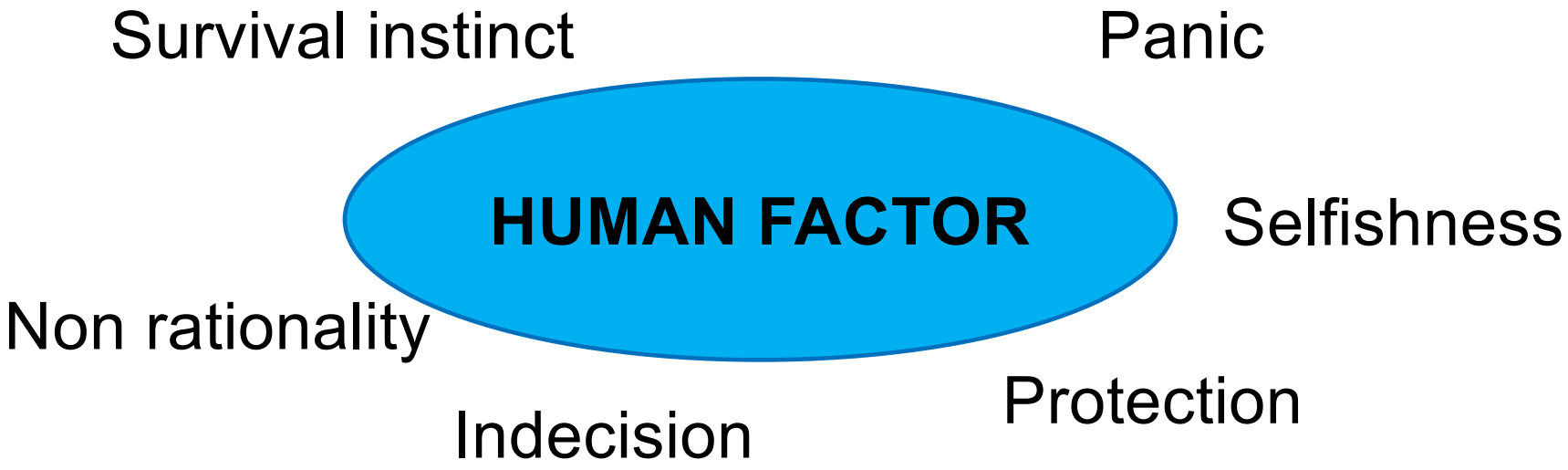
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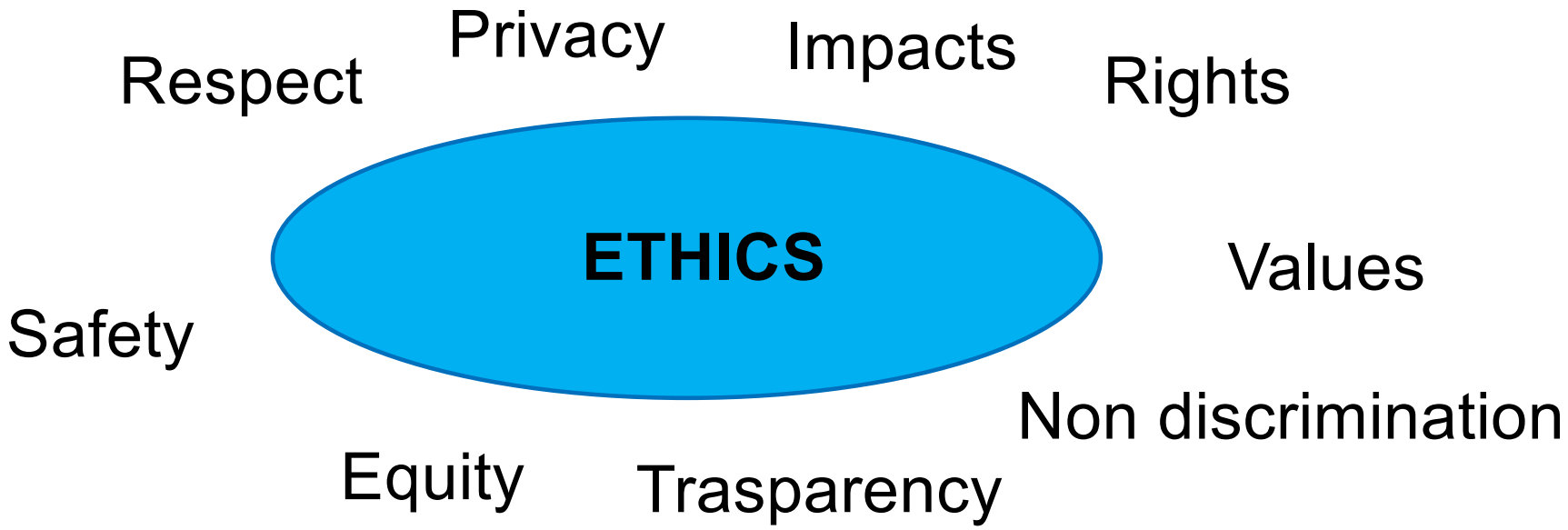
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Human factor



Plans should model the human aspects

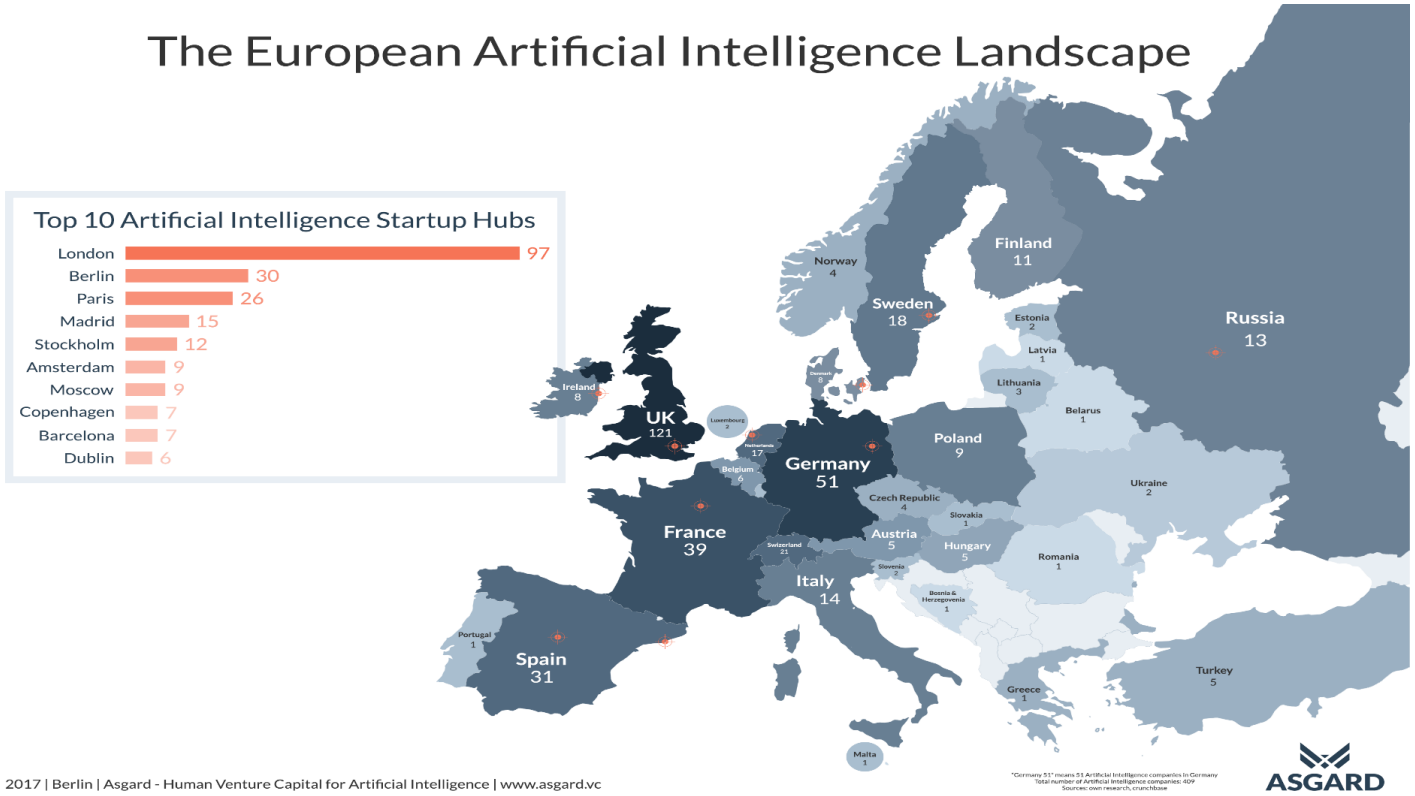
Ethical factor



Plans should model the ethical aspects

Are ethical values universal?
Are ethical principles easily and clearly defined?

EU landscape



Where AI

- European Association for Artificial Intelligence (EurAI), (previously ECCAI). Founded in 1982, is a scientific umbrella association organizing ECAI European Conference on Artificial Intelligence.

- Association for the Advancement of Artificial Intelligence (AAAI) (previously American Association for Artificial Intelligence). Founded in 1979, is a scientific association organizing the AAAI Conference on Artificial Intelligence.

- Associazione Italiana per l'Intelligenza Artificiale (AI*IA). Founded in 1988, is a scientific italian association and organizing the Conferenza Italiana di IA.

- ECAI 2025 organized in Bologna.